Week 5 – OSPF Multi-Area & EIGRP Comparison – FINAL COMPLETE

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Part A: OSPF Multi-Area (Q38-41)

38-39 R-EDGE (ABR) Config

Commands: interface g0/1 ip 10.0.1.1 255.255.255.252 no shut ; router ospf 1 network 10.0.1.0 0.0.0.3 area 1

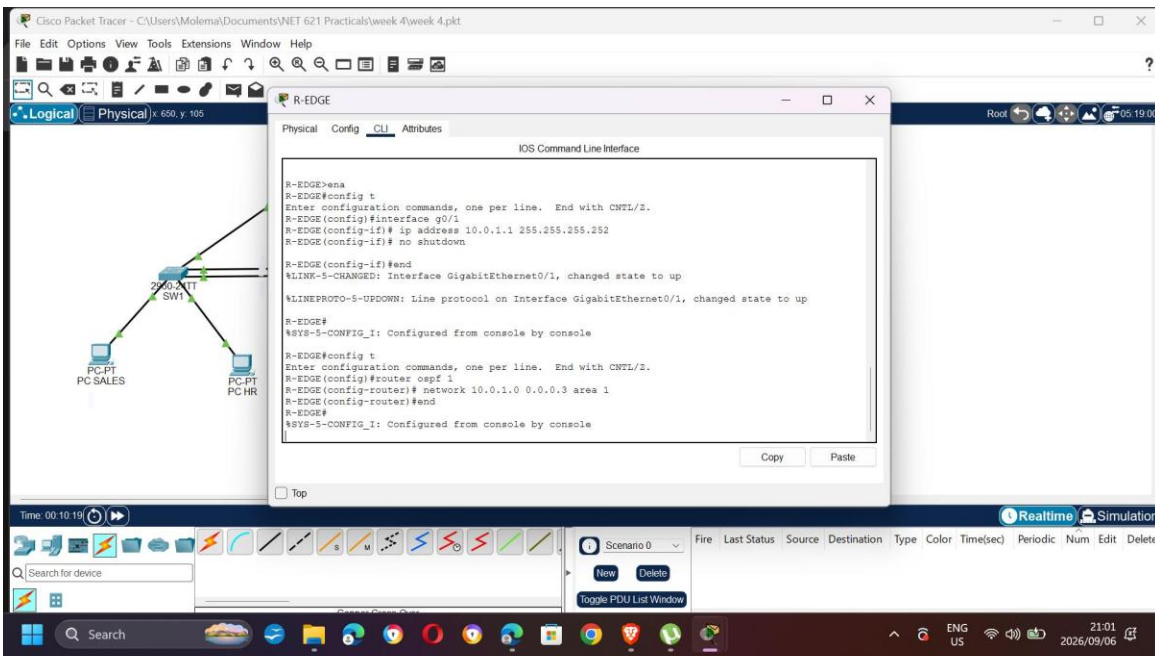

Figure 1: R-EDGE g0/1 up, OSPF Area 1 added. R-EDGE becomes ABR.

40 R3 Config + Adjacency

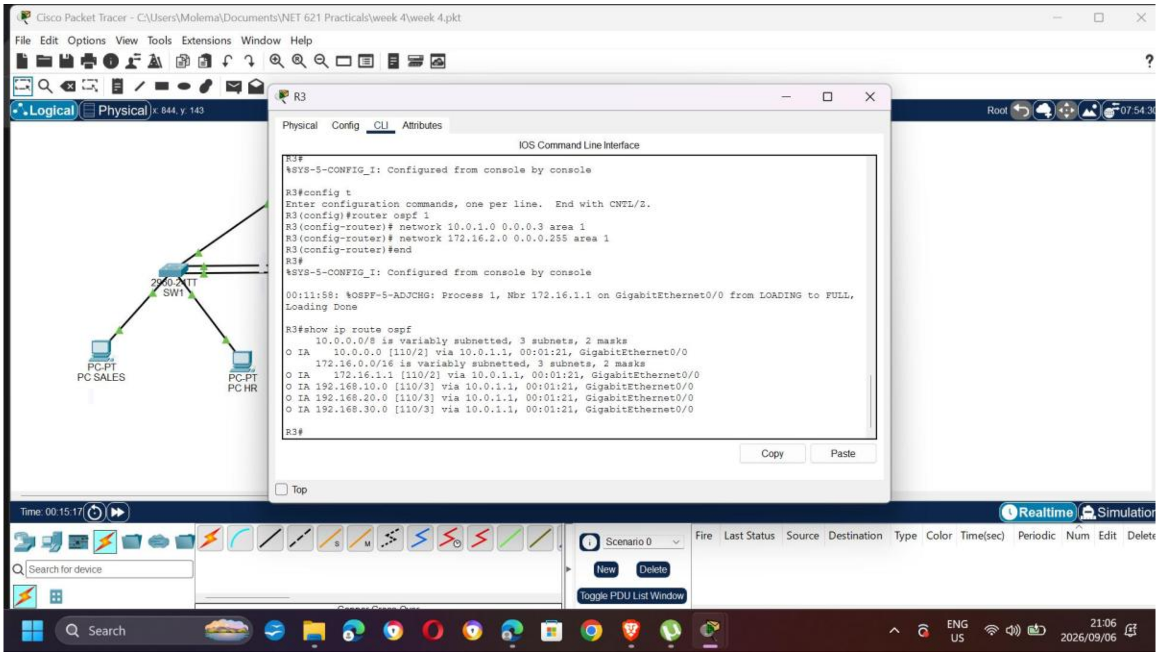

Figure 2: R3 network 10.0.1.0 area 1 + 172.16.2.0 area 1. Log OSPF-5-ADJCHG Nbr 172.16.1.1 FULL confirms adjacency to R-EDGE. show ip route ospf shows O IA 10.0.0.0, 172.16.1.1, 192.168.10/20/30.0 via 10.0.1.1

41 Verification O IA + Neighbor

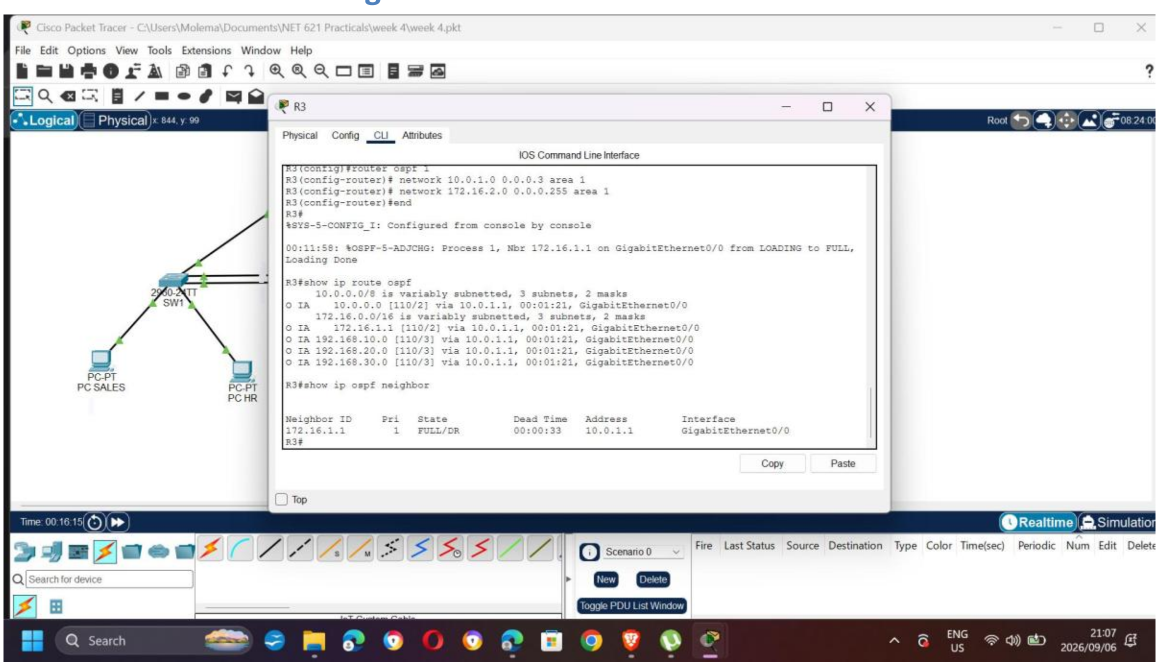

Figure 3: R3 show ip route ospf - all VLANs from Area 0 appear as O IA 192.168.10.0 [110/3], 20.0, 30.0 - inter-area. show ip ospf neighbor FULL/DR 172.16.1.1 10.0.1.1 Gig0/0 - ABR adjacency.

End-to-End Multi-Area Ping

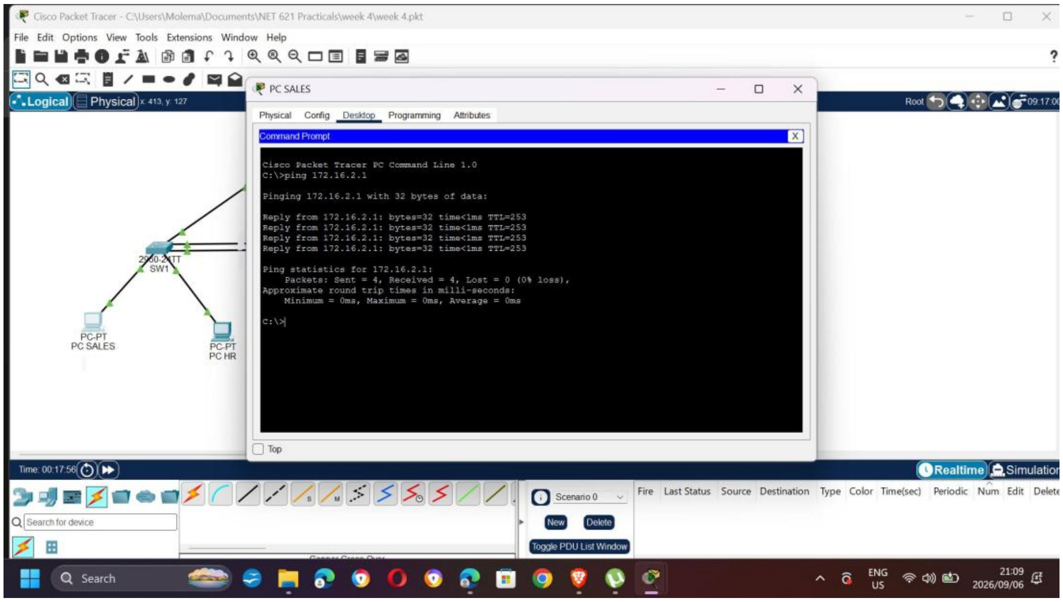

Figure 4: PC SALES (VLAN10) ping 172.16.2.1 (R3 Loopback LAN Area 1) - Reply TTL=253 (2 hops R1->R-EDGE->R3), 4 received 0% loss. Proves multi-area OSPF.

Part B: EIGRP Comparison (Q42-45)

42 EIGRP Triangle Config

Topology: R-A G0/0 10.1.12.1 <-> R-B G0/0 10.1.12.2 ; R-A G0/1 10.1.13.1 <-> R-C G0/0 10.1.13.2 ; R-B G0/1 10.1.23.1 <-> R-C G0/1 10.1.23.2 ; Loopbacks R-A 192.168.1.1/24, R-B 192.168.2.1/24, R-C 192.168.3.1/24 ; Cable Copper Cross Over
Config per router: router eigrp 100 ; network 10.0.0.0 and 192.168.x.0 exact wildcards ; no auto-summary ; metric weights 0 1 0 1 0 0 (fix K-value mismatch)

43 Summarization

On R-A: interface g0/0 ip summary-address eigrp 100 192.168.0.0 255.255.0.0 ; interface g0/1 same. Creates summary route 192.168.0.0/16 Null0 and reduces advertisements.

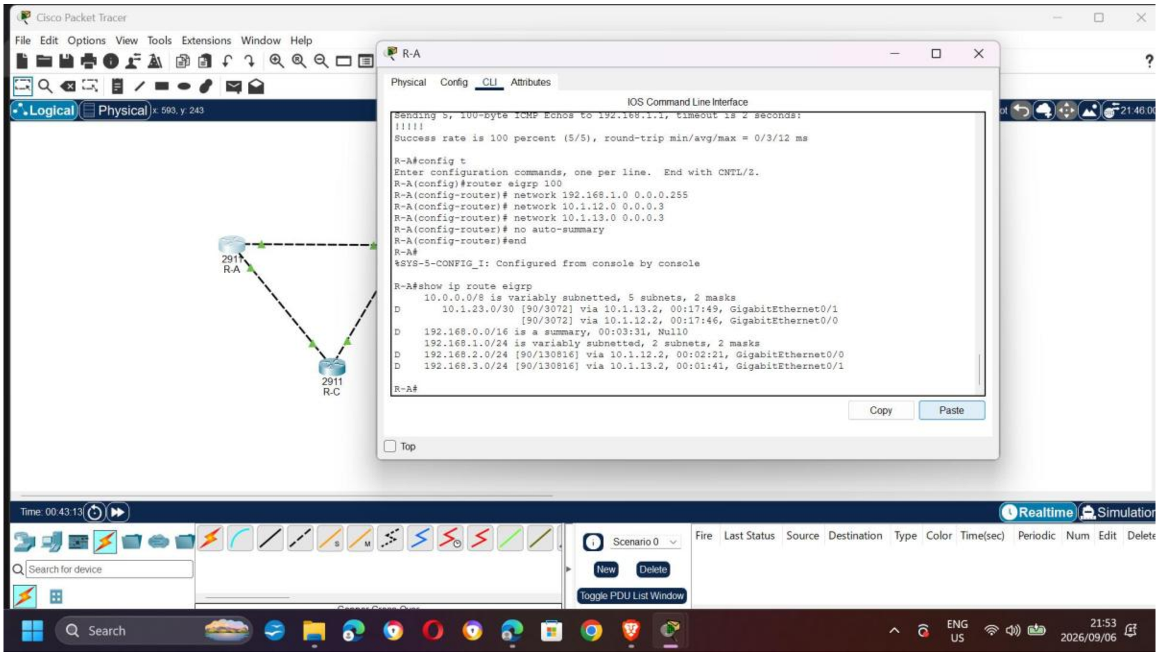

Figure 5: R-A show ip route eigrp FINAL - PROOF:
D 10.1.23.0/30 [90/3072] via 10.1.13.2 and 10.1.12.2 (load balanced triangle)
D 192.168.0.0/16 is a summary, 00:03:31, Null0 (Task 43 summarization)
D 192.168.2.0/24 [90/130816] via 10.1.12.2 (R-B LAN)
D 192.168.3.0/24 [90/130816] via 10.1.13.2 (R-C LAN)
This satisfies Q45 - EIGRP D routes with summary.

44 Failure Test (Q47)

Procedure: On R-A start extended ping to 192.168.3.1 repeat 1000 (using Protocol ip > Target > Repeat count). While pinging, shutdown g0/1 on R-A (one path). Observe 0-1 packet loss due to feasible successor via R-B. Then no shutdown to recover. OSPF equivalent test in Week 4 took longer due to LSA flood + SPF. Screenshot your continuous ping for evidence.

46 & 47 Reflection Answers

Q46 O vs O IA: O = intra-area (Type1/2 LSAs, full topology, SPF exact). O IA = inter-area (Type3 Summary LSA from ABR R-EDGE, only know ABR says reachable, not full remote topology). Benefits hierarchy, reduces flooding. In lab VLANs show O on R1 (same Area 0) but O IA on R3 (different Area 1) as seen in Figure 3.

Q47 EIGRP vs OSPF reconvergence: EIGRP reconverged faster (0-2 timeouts) in triangle due to DUAL feasible successor precomputed in topology table, local failover, no flood. OSPF slower (hello 10 dead 40 default + LSA flood to all routers in area + SPF recalc). With BFD/tuned timers OSPF can be sub-second, but out-of-box EIGRP wins for local link failure with backup path.